

Condensate Partition Coefficient Pipeline

Instruction Manual

Automated 3D segmentation of nuclear biomolecular condensates
and measurement of the partition coefficient from confocal Z-stacks

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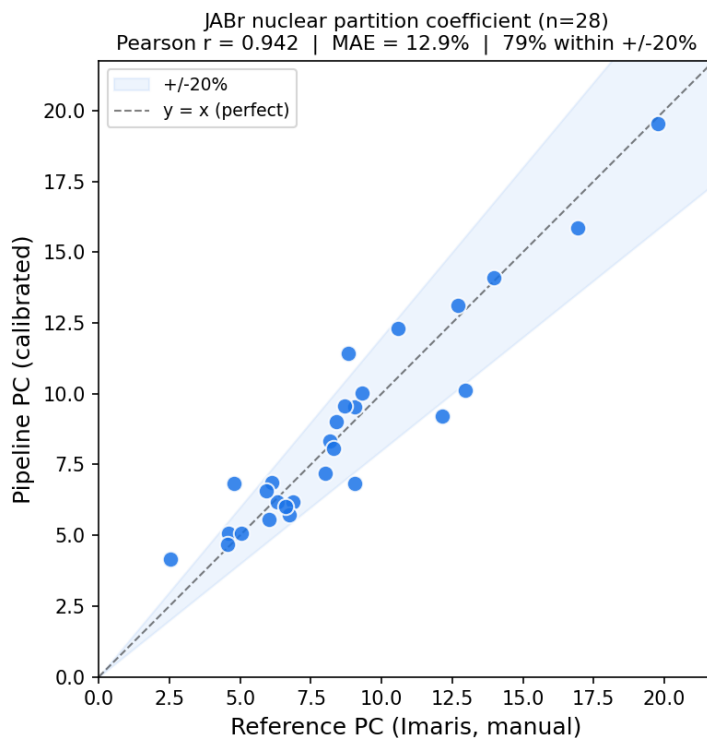
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1 Overview

This pipeline replaces the manual Imaris workflow for measuring the **partition coefficient (PC)** of nuclear biomolecular condensates. Given a two-channel confocal Z-stack, it segments nuclei and condensates in 3D, computes the PC, and reports a calibrated value on the manual reference scale.

Inputs	A two-channel Z-stack: channel 0 = nuclei, channel 1 = condensate.
Outputs	Nuclear & cytoplasmic PC, 3D masks, per-object volumes, a summary figure.
Validated on	JABr: Pearson $r = 0.942$, MAE 12.9%, 79% of cells within $\pm 20\%$ ($n = 28$).
Interfaces	Graphical (<code>run_gui.py</code>) or command line (<code>pipeline.py</code>).



2 Installation (one time)

1. Install **Python 3.10 or 3.11**.
2. Install **PyTorch** for your machine from pytorch.org (choose CPU or your CUDA version).
Example for CUDA 12.1:

```
pip install torch --index-url https://download.pytorch.org/whl/cu121
```

3. Install the remaining dependencies:

```
pip install -r requirements.txt
```

On first run, Cellpose downloads its model weights automatically (one-time ~ 100 MB). A GPU is recommended (~ 10 – 15 s per cell); CPU works but is slower.

3 Preparing your image files

The pipeline accepts either one multi-channel TIF (both channels in one file) or two separate single-channel TIFs. **Channel order matters:** channel 0 must be nuclei, channel 1 the condensate. Most axis orderings (ZCYX, CZYX, ...) are auto-detected. If results look wrong, swapped channels are the most common cause.

4 Running with the graphical interface

Launch with `python run_gui.py`. A window with **Single File** and **Batch** tabs opens.

4.1 Step by step (single cell)

1. On the **Single File** tab, choose the input mode (one multi-channel TIF, or separate files) and **Browse...** to your data.
2. Choose an **Output** folder (optional).
3. In **Settings**, set **Construct** = **JABr**. For JABr, leave all other settings at their defaults.
4. Click ► **Run Pipeline**. Progress streams in the **Output Log**; the status reads **Done** when finished.
5. Open your output folder for the masks, tables, and `results.png`.

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Single File Batch

☒ Single multi-channel TIF (OME / Cut ROI)
☐ Separate condensate + nuclei files

Multi-Channel TIF
TIF file: C:\storage\code\condensate-partition-pipeline\sam Browse...

Output
Output folder: my_output Browse...

Top-X% brightest voxels for cond. density: 75 (default 75 — recommended)
Nuclei cell probability threshold: -2.0 (default -2.0)
Construct: JABr (picks calibration + auto-routes detector)
Condensate detector: auto (auto: JABr→blob_log, else Cellpose)
Cellpose cond. model path: (blank = built-in cyto3; recommended: V3 epoch 35)
blob_log threshold: 0.03 (default 0.03 — only used when detector=blob_log)
☐ Disable GPU (use on laptop / no CUDA)

▶ Run Pipeline

Output Log



The Single-File tab configured for a JABr run: a multi-channel TIF loaded, Construct = JABr.

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Output

Output folder: my_output Browse...

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 Construct: JABr (picks calibration + auto-routes detector)
 Condensate detector: auto (auto: JABr→blob_log, else Cellpose)
 Cellpose cond. model path: (blank = built-in cyto3; recommended: V3 epoch 35)
 blob_log threshold: 0.03 (default 0.03 — only used when detector=blob_log)
☐ Disable GPU (use on laptop / no CUDA)

▶ Run Pipeline Done ✓

Output Log

```

Cond density : 319.46
Dilute density : 65.64
PC calibrated : 4.813 (JABr)
All outputs saved to: my_output
  
```

After the run: the Output Log reports the calibrated partition coefficient and the status reads Done.

5 Understanding the settings

For routine JABr analysis you only need to set **Construct**; the rest have sensible defaults.

Setting	Default	What it does
Construct	(none)	Pick JABr for the validated workflow. Auto-selects the <code>blob_log</code> detector <i>and</i> the JABr calibration so the reported PC is on the manual scale.
Condensate detector	auto	auto routes JABr→ <code>blob_log</code> , else Cellpose. Normally left on auto.

Setting	Default	What it does
Top-X% brightest voxels	75	Defines condensed-phase density; trims the dim mask halo to match the manual reference. Leave at 75.
Nuclei cell-prob. threshold	-2.0	How aggressively nuclei are detected. -2.0 suits the lab's nuclear stain.
blob_log threshold	0.03	Condensate spot-detection sensitivity (used when detector = blob_log). Leave at 0.03 for JABr.
Cellpose cond. model path	(blank)	Advanced; a fine-tuned condensate model. Leave blank for JABr.
Disable GPU	off	Tick on a laptop with no NVIDIA GPU.

The one rule for routine use: set **Construct = JABr** and run. Everything else is for experimentation or other constructs.

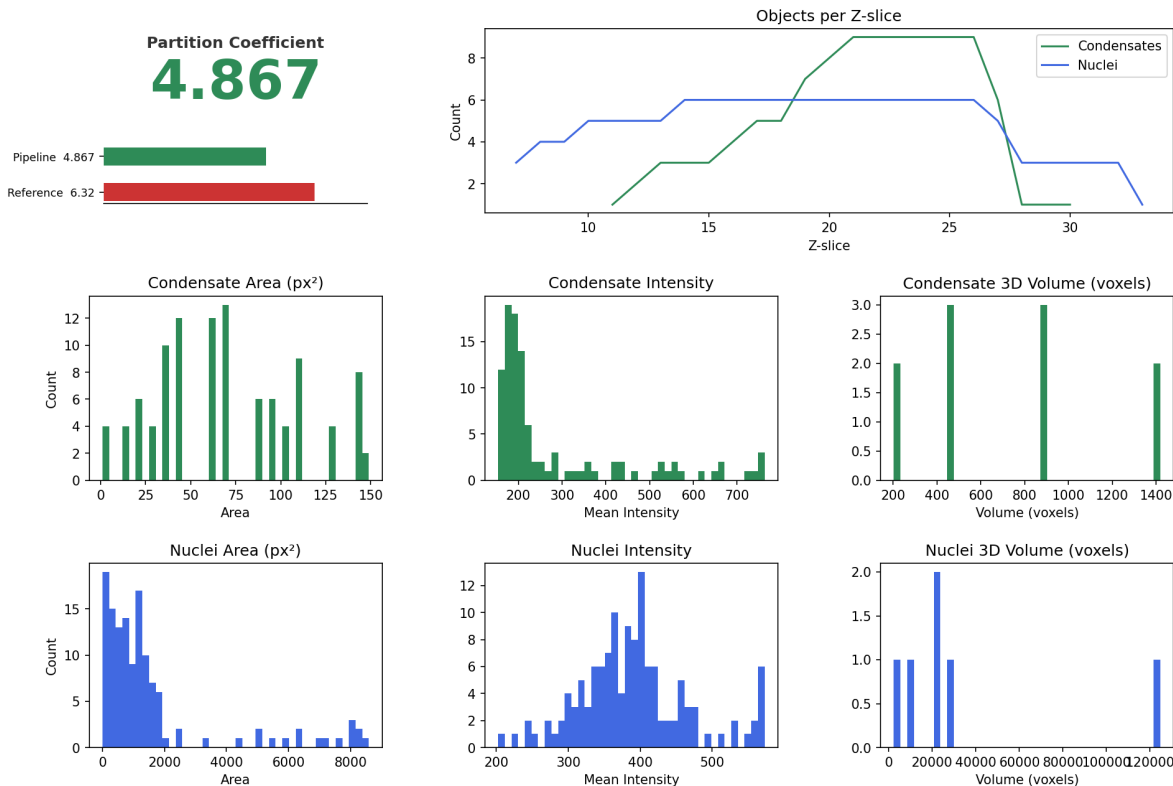
6 Reading the outputs

Every run writes: `summary.csv` (headline numbers incl. the calibrated PC), `results.png` (summary figure), `condensate_masks.tif` and `nuclei_masks.tif` (3D labeled masks), and per-object/per-slice CSVs.

For the bundled sample `JABr_Sample2_5_3.tif`, a real run produces:

```
[nuclear] PC raw 4.867 → calibrated 4.813 (manual reference 4.558, +5.6%)
```

Report the calibrated nuclear PC (the raw value is systematically $\sim 3\times$ the manual scale by construction; calibration standardizes it). **Always glance at `condensate_masks.tif`** over the raw channel in Fiji/ImageJ — a 10-second visual check catches the rare misfire.



results.png from the bundled-sample run.

7 Batch mode

On the **Batch** tab, select a folder of .tif files; every file is processed. Optionally supply the manual Imaris nuclear-PC CSV as a **Reference CSV** — the pipeline then writes **comparison.csv** (pipeline vs manual, per cell) and **scatter.png** with the correlation, RMSE, and MAE. Set **Construct = JABr** and run.

8 Command-line use

```
python pipeline.py --roi sample_data/JABr_Sample2_5_3.tif --construct JABr --output my_output
```

Separate channels: `--nuc nuclei.tif --cond condensate.tif`. Physical volumes in μm^3 : add `--voxel-xy 0.065 --voxel-z 0.3`. Force CPU: `--no-gpu`. For many files (and optional comparison against a manual reference CSV), use the GUI **Batch** tab.

9 Troubleshooting

Symptom	Likely cause / fix
PC wildly off / nuclei look empty	Channels swapped: confirm <code>ch0</code> = nuclei, <code>ch1</code> = condensate.
“0 condensates detected”	Not JABr, or very dim signal. Lower <code>blob_log</code> threshold (e.g. 0.02); confirm the raw channel has spots.
Very slow (minutes per cell)	Running on CPU. Install a CUDA PyTorch build; ensure Disable GPU is unchecked.
CUDA out of memory	Tick Disable GPU , or close other GPU programs.
Cellpose download fails	First run needs internet for model weights.
One cell’s calibrated PC looks off	Calibration is population-level; individual cells vary ($\pm 20\%$ for $\sim 80\%$ of cells). Verify against the mask.

10 Method summary

The PC is the ratio of condensed-phase to dilute-phase density (Fabrini et al.), both camera-background-subtracted. Condensed density uses the brightest 75% of voxels inside ($\text{condensate} \cap \text{nucleus}$); dilute density is the mean of the 50 lowest-intensity $10 \times 10 \times 10$ patches in the nuclear dilute region. Nuclei are segmented with Cellpose 3 (`cyto3`, 3D), cleaned, and void-filled; condensates are detected with Laplacian-of-Gaussian `blob_log` and kept when $\geq 50\%$ of the rendered sphere lies inside a nucleus. A per-construct linear (JABr) or isotonic calibration standardizes the automated PC onto the manual scale. Full rationale and the cross-construct scope/limits are in `docs/METHODS.md`.

Scope. Validated and production-ready for **JABr**. The detector is construct-specific; a leave-one-construct-out test confirmed a single detector does not generalize zero-shot (it under-detects on unseen constructs). Extending to other constructs needs per-construct calibration or few-shot retraining — see `docs/METHODS.md`, Section 5.